
Web Project Report

for

Attendance and Leave Management Portal

Version 1.0

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BS SSDP

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1. Introduction

The Attendance and Student Leave Management Portal is an exhaustive portal system. The system's role is to store the students' attendance data and help teachers when they apply for leave. I propose to develop a portal for the management of attendance and students' absences to increase data storage efficiency and accessibility. To manage and record the attendance of students effectively, this portal should provide a solution for teachers

2. Problem Statement

In Higher educational institutions, managing the attendance of students and reviewing Students applications are the two tasks that take most of the time. Teachers and staff associated with portals are affected by the outdated methods and technology used, which result in errors, time-consuming load times, poor user interfaces, and a lack of accessibility to users. Modern techniques and modern software should be used to modernize the portal to make it more user-friendly

3. Project Selection Rationale

I chose to develop Student Attendance and Leave Management portal because portals are affected by the outdated methods and technology used, which result in errors, and time-consuming load times. By using modern database techniques to make it more reliable

This project is of great relevance to my studies, since I am a fourth semester student in IT. This project not only help me in education sector but also play an important part in growth of technical and problem-solving skill

4. Tools and Technologies Used

Frontend:

1. **HTML:** For the structure of the web pages.
2. **CSS:** For styling and layout.
3. **JavaScript:** For interactivity and client-side validation.

Backend & Database:

1. **PHP:** For server-side scripting and logic.
2. **MySQL:** For database management and storage.

Server Environment:

1. **XAMPP:** Used as the local server environment to test and deploy the application.

5. Targeted Audience

Primary Audience:

1. Teachers
2. School Administrators
3. IT Staff in Educational Institutions

Importance and Meeting Their Needs:

Teachers:

Importance: Directly handle attendance and leave management.

Needs Met: Efficient attendance recording, user-friendly interface, real-time access to data.

School Administrators:

Importance: Oversee school operations and ensure accurate record-keeping.

Needs Met: Reliable attendance data, comprehensive reports, improved communication.

IT Staff:

Importance: Implement and maintain school software systems.

Needs Met: Easy implementation and maintenance, minimal technical support required, scalable solution.

6. Effectiveness of the Project

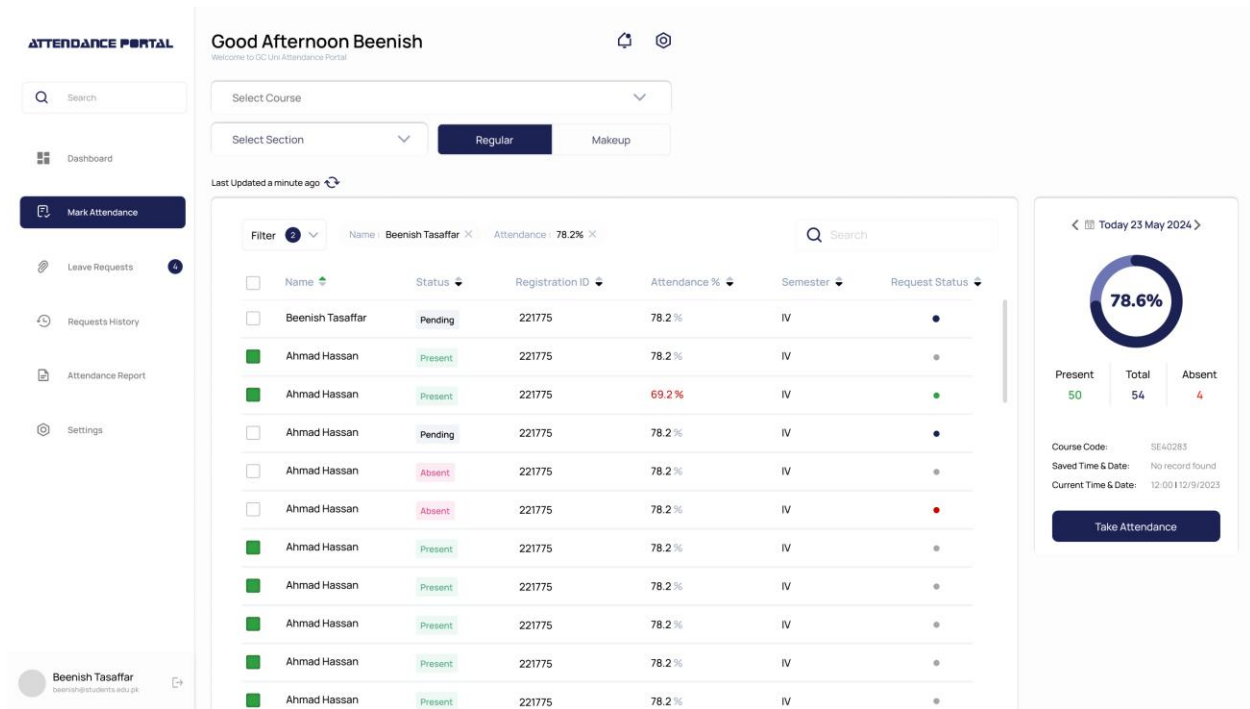
Anticipated Effectiveness:

- **Impact:** The Attendance and Student Leave Management Portal is expected to significantly reduce the time and effort required for managing attendance and leave applications, leading to more efficient administrative processes
- **Adoption:** High potential for widespread adoption in educational institutions due to its user-friendly interface and comprehensive features
- **User Satisfaction:** By addressing the specific needs of teachers, administrators, and IT staff, the portal aims to improve overall satisfaction and ease of use
- **Data Accuracy:** Enhanced accuracy in attendance records and leave management, reducing errors

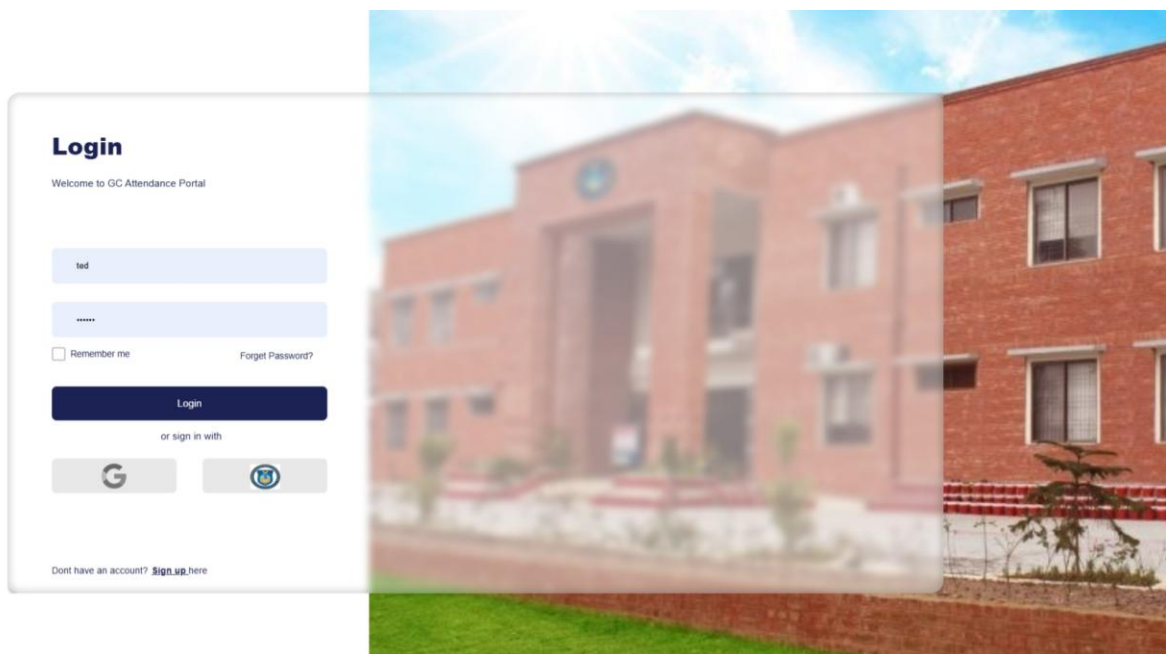
Measuring Success:

- **User Feedback:** Collect feedback from teachers and administrators about their experience using the portal. Conduct simple surveys to gather opinions on ease of use, functionality, and overall satisfaction
- **Usage Metrics:** Track the number of users actively using the portal. Monitor the frequency of logins and the number of attendance records and leave applications processed
- **Adoption Rates:** Measure how many educational institutions and individual users have adopted the portal
- **Error Reduction:** Compare the number of errors in attendance and leave management before and after implementing the portal
- **Support Requests:** Track the number of support requests and issues reported by users to identify areas for improvement

7. Figma Design:



8. Frontend:



After Logging In:

The screenshot shows the 'Attendance Portal' interface. On the left is a sidebar with navigation links: Search, Dashboard, Mark Attendance (highlighted), Leave Requests, Requests History, Attendance Report, and Settings. At the bottom of the sidebar is a 'Log out' button. The main content area is titled 'Good Afternoon CS001' and 'Welcome to GC Uni Attendance Portal'. It features a header with 'Select Course' and 'Select Section' dropdowns, and buttons for 'Regular', 'Makeup', and 'Update'. Below this is a 'Search List' input field and a table with columns: Name, Registration ID, Attendance%, Semester, and Request Status. The table is currently empty, displaying 'No Rows To Show'. On the right side, there is a summary card for 'Today Wed Feb 11 2026' showing a 0% attendance rate. It includes a circular progress indicator and a table with 'Present' (0), 'Total' (0), and 'Absent' (0). Below this, it shows 'Course Code: SE40283', 'Saved Time: No record found', and 'Current Time: 18:35 | 2/11/2026'. A 'Mark Attendance' button is at the bottom of the summary card.

Only those courses will be shown, which don't have their attendance marked:

This screenshot is similar to the previous one, but the 'Select Course' dropdown menu is open, showing a list of courses. The first option is 'CS101 - Introduction to Programming'. The rest of the interface, including the sidebar, main content area with the empty table, and the right-side summary card, remains the same.

After Selecting it:

Attendance Portal

Search

Dashboard

Mark Attendance

Leave Requests

Requests History

Attendance Report

Settings

Log out

Good Afternoon CS001

Welcome to GC Uni Attendance Portal

CSE101 - Introduction to Programming

Select Section

Regular

Makeup

Update

Select Course

CSE101 - Introduction to Programming

Search List

Name	Registration ID	Attendance%	Semester	Request Status
Ahmad Hassan	221775	100.00%	0	Approved
Beenish Tasaffar	221847	100.00%	0	No Request
Faraz Ashraf	221899	50.00%	0	No Request

Log out

Today Wed Feb 11 2026

0.0%

Present

0

Total

3

Absent

3

Course Code:

CSE101

Saved Time:

No record found

Current Time:

18:36 | 2/11/2026

Mark Attendance

Live Stats showing on the Sidebar:

Attendance Portal

Search

Dashboard

Mark Attendance

Leave Requests

Requests History

Attendance Report

Settings

Log out

Good Afternoon CS001

Welcome to GC Uni Attendance Portal

CSE101 - Introduction to Programming

Select Section

Regular

Makeup

Update

Select Course

CSE101 - Introduction to Programming

Search List

Name	Registration ID	Attendance%	Semester	Request Status
Ahmad Hassan	221775	100.00%	0	Approved
Beenish Tasaffar	221847	100.00%	0	No Request
Faraz Ashraf	221899	50.00%	0	No Request

Log out

Today Wed Feb 11 2026

66.7%

Present

2

Total

3

Absent

1

Course Code:

CSE101

Saved Time:

No record found

Current Time:

18:37 | 2/11/2026

Mark Attendance

After Pressing Mark Attendance:

Attendance Portal

Search

Dashboard

Mark Attendance

Leave Requests

Requests History

Attendance Report

Settings

Log out

Good Afternoon CS001

Welcome to GC Uni Attendance Portal

CSE101 - Introduction to Programming

Select Section

Search List

Name	Registration ID	Attendance%	Semester	Request Status
Ahmad Hassan	221775	100.00	0	Approved
Beenish Tasaffar	221847	100.00	0	No Request
Faraz Ashraf	221899	50.00%	0	No Request

localhost says

Attendance Marked Successfully

OK

Today Wed Feb 11 2026

66.7%

Present 2

Total 3

Absent 1

Course Code: CSE101

Saved Time: No record found

Current Time: 18:37 | 2/11/2026

Mark Attendance

Database:

Before:

After:

			att_id	course_id	th_regid	att_status	att_date	att_timerecorded	att_type	stu_regld	leave_id
☐	Edit	Copy	Delete	329	CSE101	CS001	Absent	2026-05-20 09:46:32	Makeup	221775	NULL
☐	Edit	Copy	Delete	330	CSE101	CS001	Present	2026-05-20 09:46:32	Makeup	221847	NULL
☐	Edit	Copy	Delete	331	CSE101	CS001	Absent	2026-05-20 09:46:32	Makeup	221899	NULL
☐	Edit	Copy	Delete	332	CSE101	CS001	Present	2026-05-20 09:46:32	Makeup	221775	NULL
☐	Edit	Copy	Delete	333	CSE101	CS001	Present	2026-02-10 22:12:32	Regular	221775	NULL
☐	Edit	Copy	Delete	334	CSE101	CS001	Present	2026-02-10 22:12:32	Regular	221847	NULL
☐	Edit	Copy	Delete	335	CSE101	CS001	Present	2026-02-10 22:12:32	Regular	221899	NULL
☐	Edit	Copy	Delete	339	CSE101	CS001	Present	2026-02-11 18:38:54	Regular	221775	1
☐	Edit	Copy	Delete	340	CSE101	CS001	Present	2026-02-11 18:38:54	Regular	221847	NULL
☐	Edit	Copy	Delete	341	CSE101	CS001	Absent	2026-02-11 18:38:54	Regular	221899	NULL

☐ Check all With selected: Edit Copy Delete Export

File Structure:

```
C:\Windows\System32\cmd.e  X  +  v

T:\Programs\xampp\htdocs\DBProject_Attendance_Portal>tree/f
Folder PATH listing for volume Ted
Volume serial number is 8C49-D994
T:.
  api.php
  attendance_portal_beenishtasaffar.sql
  db_connect.php
  index.php
  login.php
  logout.php
  portal.php
  register.php
  test_db.php
  web project demo.mp4
  --css
  |   App.css
  |   login.css
  |   portal.css
  |   register.css
  --images
  |   google_logo.png
  |   login_background.jpg
  |   logo.png
  |   unilogo.jpg
T:\Programs\xampp\htdocs\DBProject_Attendance_Portal>
```

9. Backend Code:

```
<?php

include 'db_connect.php';

if ($_SERVER['REQUEST_METHOD'] === 'OPTIONS') {
    http_response_code(200);
    exit();
}

$action = $_GET['action'] ?? '';

switch ($action) {
    case 'login':
        handleLogin($conn);
        break;
    case 'register':
        handleRegister($conn);
        break;
    case 'getStudents':
        handleGetStudents($conn);
        break;
    case 'getTeacherCourses':
        handleGetTeacherCourses($conn);
        break;
    case 'markAttendance':
        handleMarkAttendance($conn);
        break;
    case 'getPendingLeaves':
        handleGetPendingLeaves($conn);
        break;
    case 'updateLeaveStatus':
        handleUpdateLeaveStatus($conn);
        break;
    default:
        echo json_encode(["message" => "Invalid API Action",
"received_action" => $action]);
        break;
}

$conn->close();

function handleLogin($conn) {
    $data = json_decode(file_get_contents("php://input"));

    if(isset($data->LoginUserName) && isset($data->LoginPassword)) {
        $stmt = $conn->prepare("SELECT th_regId FROM login WHERE l_username =
? AND l_password = ?");
```

```

$stmt->bind_param("ss", $data->LoginUserName, $data->LoginPassword);
$stmt->execute();
$result = $stmt->get_result();

if ($result->num_rows > 0) {
    $rows = [];
    while($row = $result->fetch_assoc()) {
        $rows[] = $row;
    }
    echo json_encode($rows);
} else {
    echo json_encode(["message" => "Credentials Dont Match"]);
}
$stmt->close();
} else {
    echo json_encode(["message" => "Invalid Input"]);
}
}

function handleRegister($conn) {
    $data = json_decode(file_get_contents("php://input"));

    if(isset($data->Email)) {
        $stmt = $conn->prepare("INSERT INTO login(th_regId, l_username,
l_password, l_recmail) VALUES (?, ?, ?, ?)");
        $stmt->bind_param("ssss", $data->UserRegId, $data->UserName, $data->
Password, $data->Email);

        if ($stmt->execute()) {
            echo json_encode(["message" => "User Added"]);
        } else {
            echo json_encode(["error" => "Registration Failed: " . $stmt->
error]);
        }
        $stmt->close();
    } else {
        echo json_encode(["error" => "Missing Email"]);
    }
}

function handleGetStudents($conn) {
    $course_id = $_GET['course_id'] ?? '';
    $leave_date = $_GET['leave_date'] ?? '';
    $th_regId = $_GET['th_regId'] ?? '';

    $sql = "SELECT
            students.stu_name,
            sub_query.IDs AS stuRegIds,
            sub_query.attendance_percentage,
            2026 - students.stu_batch AS semester,

```

```

        sub_query.RQs AS requestStatus,
        sub_query.lids AS leave_id
    FROM students
    LEFT JOIN (
        SELECT
            studentregisteredcourses.stu_regId AS IDs,
            ROUND(
                COUNT(DISTINCT CASE WHEN attendance.att_status =
'Present' THEN attendance.att_date END) * 100.0
                / GREATEST(COUNT(DISTINCT attendance.att_date), 1)
                , 2) AS attendance_percentage,
            COALESCE(MAX(leave_application.request_status), 'No
Request') AS RQs,
            MAX(leave_application.leave_id) AS lids
        FROM studentregisteredcourses
        INNER JOIN students ON studentregisteredcourses.stu_regId =
students.stu_regId
        LEFT JOIN attendance ON studentregisteredcourses.stu_regId =
attendance.stu_regId
        AND studentregisteredcourses.course_id =
attendance.course_id
        AND attendance.th_regid = ?
        LEFT JOIN leave_application ON students.stu_regId =
leave_application.stu_regId
        AND leave_application.leave_date = ?
        WHERE studentregisteredcourses.course_id = ?
        GROUP BY studentregisteredcourses.stu_regId
    ) AS sub_query ON students.stu_regId = sub_query.IDs
    WHERE sub_query.IDs IS NOT NULL";

$stmt = $conn->prepare($sql);

$stmt->bind_param("sss", $th_regId, $leave_date, $course_id);

if ($stmt->execute()) {
    $result = $stmt->get_result();
    $data = [];
    while ($row = $result->fetch_assoc()) {
        $data[] = $row;
    }
    echo json_encode($data);
} else {
    echo json_encode(["error" => "Query Failed: " . $stmt->error]);
}
$stmt->close();
}

function handleGetTeacherCourses($conn) {
    $th_regId = $_GET['th_regId'] ?? '';
    $att_date = $_GET['att_date'] ?? '';

```

```

    $sql = "SELECT courses.course_id AS courseIDS, CONCAT(courses.course_id,
' - ', courses.course_name) AS courseOption
        FROM (
            SELECT trc.course_id, crs.course_name
            FROM teacherregisteredcourses trc
            JOIN courses crs ON trc.course_id = crs.course_id
            WHERE trc.th_regId = ?
        ) AS courses
        LEFT JOIN attendance at ON courses.course_id = at.course_id AND
at.att_date = ?
        WHERE at.course_id IS NULL";

$stmt = $conn->prepare($sql);
$stmt->bind_param("ss", $th_regId, $att_date);

if ($stmt->execute()) {
    $result = $stmt->get_result();
    $data = [];
    while ($row = $result->fetch_assoc()) {
        $data[] = $row;
    }
    echo json_encode($data);
} else {
    echo json_encode(["error" => "Query Failed: " . $stmt->error]);
}
$stmt->close();
}

function handleMarkAttendance($conn) {

    $input = json_decode(file_get_contents("php://input"), true);
    $data = $input['data'] ?? null;

    if (!is_array($data)) {
        http_response_code(400);
        die(json_encode(["error" => "Invalid data format"]));
    }

    $conn->begin_transaction();

    try {

        $stmtInsert = $conn->prepare("INSERT INTO attendance (course_id,
th_regid, att_status, att_date, att_timerecorded, att_type, stu_regId,
leave_id) VALUES (?, ?, ?, ?, ?, ?, ?, ?)");

        foreach ($data as $item) {
            $stmtInsert->bind_param("sssssssi",
                $item['course_id'],

```

```

        $item['th_regId'],
        $item['isSelected'],
        $item['date'],
        $item['time'],
        $item['Attendance_type'],
        $item['stuRegIds'],
        $item['leave_id']
    );
    if (!$stmtInsert->execute()) {
        throw new Exception("Insert Attendance Failed: " .
$stmtInsert->error());
    }
}
$stmtInsert->close();

$uniqueCourses = [];
foreach ($data as $item) {
    $key = $item['th_regId'] . '-' . $item['course_id'];
    $uniqueCourses[$key] = [
        'th_regId' => $item['th_regId'],
        'course_id' => $item['course_id']
    ];
}

$stmtUpdateTeacher = $conn->prepare("UPDATE teacherregisteredcourses
SET trc_totalClassesTaken = trc_totalClassesTaken + 1 WHERE th_regId = ? AND
course_id = ?");
foreach ($uniqueCourses as $course) {
    $stmtUpdateTeacher->bind_param("ss", $course['th_regId'],
$course['course_id']);
    if (!$stmtUpdateTeacher->execute()) {
        throw new Exception("Update Teacher Failed: " .
$stmtUpdateTeacher->error());
    }
}
$stmtUpdateTeacher->close();

$presentStudents = [];
foreach ($data as $item) {
    if ($item['isSelected'] === 'Present') {
        $key = $item['stuRegIds'] . '-' . $item['course_id'];
        $presentStudents[$key] = [
            'stuRegIds' => $item['stuRegIds'],
            'course_id' => $item['course_id']
        ];
    }
}
}

```

```

        $stmtUpdateStudent = $conn->prepare("UPDATE studentregisteredcourses
SET src_totalPresent = src_totalPresent + 1 WHERE stu_RegId = ? AND course_id
= ?");
        foreach ($presentStudents as $student) {
            $stmtUpdateStudent->bind_param("ss", $student['stuRegIds'],
$student['course_id']);
            if (!$stmtUpdateStudent->execute()) {
                throw new Exception("Update Student Failed: " .
$stmtUpdateStudent->error());
            }
        }
        $stmtUpdateStudent->close();

        $conn->commit();
        echo json_encode(["message" => "Data inserted and incremented
successfully"]);

    } catch (Exception $e) {

        $conn->rollback();
        http_response_code(500);
        echo json_encode(["error" => "Transaction failed: " . $e-
>getMessage()]);
    }
}

function handleGetPendingLeaves($conn) {
    $th_regId = $_GET['th_regId'] ?? '';

    $sql = "SELECT
            la.leave_id,
            la.stu_regId,
            s.stu_name,
            la.leave_title,
            la.leave_message,
            la.leave_date,
            la.course_id
        FROM leave_application la
        JOIN students s ON la.stu_regId = s.stu_regId
        JOIN teacherregisteredcourses trc ON la.course_id = trc.course_id
        WHERE trc.th_regId = ? AND la.request_status = 'Pending'";

    $stmt = $conn->prepare($sql);
    $stmt->bind_param("s", $th_regId);

    if ($stmt->execute()) {
        $result = $stmt->get_result();
        $data = [];
        while ($row = $result->fetch_assoc()) {
            $data[] = $row;
        }
    }
}

```

```

    }
    echo json_encode($data);
} else {
    echo json_encode(["error" => "Query Failed: " . $stmt->error]);
}
$stmt->close();
}

function handleUpdateLeaveStatus($conn) {
    $data = json_decode(file_get_contents("php://input"));

    if (isset($data->leave_id) && isset($data->status)) {
        $stmt = $conn->prepare("UPDATE leave_application SET request_status =
? WHERE leave_id = ?");
        $stmt->bind_param("si", $data->status, $data->leave_id);

        if ($stmt->execute()) {
            echo json_encode(["message" => "Leave status updated to " .
$data->status]);
        } else {
            echo json_encode(["error" => "Update Failed: " . $stmt->error]);
        }
        $stmt->close();
    } else {
        echo json_encode(["error" => "Invalid Input"]);
    }
}
?>

```

10. Frontend code:

portal.php

```
<?php
session_start();

if (!isset($_SESSION['isLoggedIn']) || $_SESSION['isLoggedIn'] !== true) {
    header("Location: login.php");
    exit();
}

$loggedUserId = $_SESSION['LoggedUserId'];
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard - AU Attendance Portal</title>

    <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600&displ
ay=swap" rel="stylesheet">
    <link href='https://unpkg.com/boxicons@2.1.4/css/boxicons.min.css'
rel='stylesheet'>

    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/ag-grid-
community/styles/ag-grid.css">
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/ag-grid-
community/styles/ag-theme-quartz.css">

    <style>
        :root {
            --primary-color: #1D2254;
            --bg-color: #F4F7FE;
            --white: #ffffff;
            --text-grey: #A3AED0;
            --text-dark: #2B3674;
            --green: #05CD99;
            --red: #EE5D50;
        }

        * { margin: 0; padding: 0; box-sizing: border-box; font-family:
'Poppins', sans-serif; }

        body { background-color: var(--bg-color); height: 100vh; overflow:
hidden; display: flex; }

        .sidebar {
            width: 250px;
```

```

        background: var(--white);
        height: 100%;
        padding: 20px;
        display: flex;
        flex-direction: column;
        border-right: 1px solid #e0e0e0;
    }
    .logo { display: flex; align-items: center; margin-bottom: 30px; font-
weight: 700; color: var(--primary-color); font-size: 1.2rem; }
    .logo img { width: 30px; margin-right: 10px; }

    .nav-links { list-style: none; flex-grow: 1; }
    .nav-links li { margin-bottom: 8px; }
    .nav-links a {
        display: flex;
        align-items: center;
        padding: 12px 15px;
        color: var(--text-grey);
        text-decoration: none;
        border-radius: 10px;
        transition: 0.3s;
        font-size: 0.95rem;
    }
    .nav-links a i { font-size: 1.4rem; margin-right: 12px; }
    .nav-links a:hover, .nav-links a.active {
        background-color: var(--primary-color);
        color: var(--white);
    }
    .nav-links a.active i { color: var(--white); }

    .main-content {
        flex: 1;
        padding: 20px 30px;
        display: flex;
        flex-direction: column;
        overflow-y: auto;
    }

    header { display: flex; justify-content: space-between; align-items:
center; margin-bottom: 20px; }
    .header-left h2 { color: var(--text-dark); font-size: 1.5rem; }
    .header-left p { color: var(--text-grey); font-size: 0.85rem; }
    .header-right { display: flex; gap: 15px; align-items: center; }
    .icon-btn { background: var(--white); border: none; padding: 8px; border-
radius: 50%; color: var(--text-grey); cursor: pointer; box-shadow: 0 2px 5px
rgba(0,0,0,0.05); }

    .controls-bar {
        display: flex;
        background: var(--white);
        padding: 15px;
        border-radius: 15px;

```

```

        margin-bottom: 20px;
        align-items: center;
        gap: 20px;
        box-shadow: 0 5px 15px rgba(0,0,0,0.02);
    }
    select {
        padding: 10px 15px;
        border: 1px solid #e0e0e0;
        border-radius: 10px;
        color: var(--text-grey);
        outline: none;
        background: transparent;
        min-width: 150px;
    }

    .toggle-container {
        display: flex;
        background: #F4F7FE;
        padding: 4px;
        border-radius: 20px;
    }
    .toggle-btn {
        padding: 8px 20px;
        border-radius: 16px;
        cursor: pointer;
        font-size: 0.9rem;
        color: var(--text-grey);
        transition: 0.3s;
        display: flex;
        align-items: center;
    }
    .toggle-btn.active {
        background: var(--primary-color);
        color: var(--white);
    }

    input[type="radio"] { display: none; }

    .grid-container {
        background: var(--white);
        border-radius: 20px;
        padding: 20px;
        flex: 1;
        display: flex;
        flex-direction: column;
        box-shadow: 0 5px 15px rgba(0,0,0,0.02);
    }
    .grid-header { display: flex; justify-content: space-between; align-items: center; margin-bottom: 15px; }

    .ag-theme-quartz {
        --ag-header-background-color: transparent;

```

```

        --ag-header-foreground-color: var(--text-grey);
        --ag-row-hover-color: #F4F7FE;
        --ag-selected-row-background-color: #F4F7FE;
        font-family: 'Poppins', sans-serif;
        width: 100%;
        height: 100%;
    }

    .dot { height: 10px; width: 10px; border-radius: 50%; display: inline-block; margin-right: 5px; }
    .live-blue-dot { background-color: var(--primary-color); }
    .accepted-green-dot { background-color: var(--green); }
    .rejected-red-dot { background-color: var(--red); }
    .grey-dot { background-color: #ddd; }

    .right-sidebar {
        width: 300px;
        background: var(--bg-color);
        padding: 20px 20px 20px 0;
        display: flex;
        flex-direction: column;
        gap: 20px;
    }

    .summary-widget {
        background: var(--white);
        border-radius: 20px;
        padding: 20px;
        text-align: center;
        box-shadow: 0 5px 15px rgba(0,0,0,0.02);
        height: 100%;
    }

    .progress-circle {
        width: 120px; height: 120px; border-radius: 50%;
        background: conic-gradient(var(--primary-color) 0%, #edf2f7 0deg);
        display: flex; align-items: center; justify-content: center;
        margin: 0 auto 20px; position: relative;
    }

    .progress-circle-inner {
        width: 100px; height: 100px; background: white; border-radius: 50%;
        display: flex; align-items: center; justify-content: center;
        font-weight: 700; font-size: 1.2rem; color: var(--primary-color);
    }

    .stats-row { display: flex; justify-content: space-around; margin-bottom: 20px; }
    .stat-item h5 { font-size: 0.8rem; color: var(--text-grey); }
    .stat-item p { font-weight: 700; font-size: 1rem; }
    .stat-present { color: var(--green); }
    .stat-absent { color: var(--red); }

    .info-rows { text-align: left; font-size: 0.8rem; margin-bottom: 20px; }
    .info-row { display: flex; justify-content: space-between; margin-bottom: 8px; color: var(--text-grey); }

```

```
.info-val { font-weight: 600; color: var(--text-dark); }

.btn-main {
  width: 100%; background: var(--primary-color); color: white; padding:
12px;
  border: none; border-radius: 10px; font-weight: 500; cursor: pointer;
transition: 0.2s;
}
.btn-main:hover { background: #151a40; }

.update-btn { cursor: pointer; display: flex; align-items: center; color:
var(--primary-color); }

.leave-widget {
  margin-bottom: 20px;
  position: relative;
  display: flex;
  align-items: center;
  justify-content: center;
}
.leave-card {
  background: var(--white);
  border-radius: 20px;
  padding: 20px;
  box-shadow: 0 5px 15px rgba(0,0,0,0.02);
  width: 100%;
  position: relative;
  min-height: 220px;
  display: flex;
  flex-direction: column;
}
.leave-header {
  display: flex;
  justify-content: space-between;
  align-items: flex-start;
  margin-bottom: 15px;
}
.leave-stu-name { font-weight: 700; font-size: 1rem; color: var(--text-
dark); }
.leave-title { font-weight: 600; font-size: 0.9rem; color: var(--text-
dark); text-align: right; }
.leave-id-badge {
  background: #F4F7FE; padding: 2px 8px; border-radius: 5px;
  font-size: 0.75rem; color: var(--text-grey); margin-top: 5px;
display: inline-block;
}
.leave-message {
  font-size: 0.85rem; color: var(--text-grey); line-height: 1.4;
  margin-bottom: 20px; flex-grow: 1;
  overflow-y: auto; max-height: 100px;
}
.leave-footer {
```

```

        display: flex; justify-content: space-between; align-items: center;
        margin-top: auto;
    }
    .leave-date { font-size: 0.75rem; color: var(--text-grey); }
    .leave-actions { display: flex; gap: 10px; }
    .btn-accept { background: var(--green); color: white; border: none;
padding: 8px 15px; border-radius: 8px; cursor: pointer; font-size: 0.8rem;
transition: 0.2s; }
    .btn-reject { background: var(--red); color: white; border: none;
padding: 8px 15px; border-radius: 8px; cursor: pointer; font-size: 0.8rem;
transition: 0.2s; }
    .btn-accept:hover { opacity: 0.9; }
    .btn-reject:hover { opacity: 0.9; }

    .nav-arrow {
        background: var(--primary-color); color: white; border: none;
        width: 30px; height: 30px; border-radius: 50%;
        display: flex; align-items: center; justify-content: center;
        cursor: pointer; position: absolute; top: 50%; transform:
translateY(-50%);
        z-index: 10;
        box-shadow: 0 2px 5px rgba(0,0,0,0.2);
    }
    .nav-prev { left: -10px; }
    .nav-next { right: -10px; }
</style>
</head>
<body>

    <nav class="sidebar">
        <div class="logo">

            Attendance Portal
        </div>

        <ul class="nav-links">
            <li><a href="#"><i class='bx bx-search'></i> Search</a></li>
            <li><a href="#"><i class='bx bxs-dashboard'></i> Dashboard</a></li>
            <li><a href="#" class="active"><i class='bx bx-bookmark-plus'></i>
Mark Attendance</a></li>
            <li><a href="#"><i class='bx bx-paperclip'></i> Leave
Requests</a></li>
            <li><a href="#"><i class='bx bx-timer'></i> Requests History</a></li>
            <li><a href="#"><i class='bx bx-note'></i> Attendance Report</a></li>
            <li><a href="#"><i class='bx bx-cog'></i> Settings</a></li>
        </ul>

        <ul class="nav-links" style="flex-grow: 0;">
            <li><a href="login.php" class="logout"><i class='bx bx-log-out'></i>
Log out</a></li>
        </ul>
    </nav>

```

```

<div class="main-content">
  <header>
    <div class="header-left">
      <h2 id='headone'>Good Afternoon <?php echo
htmlspecialchars($loggedUserId); ?></h2>
      <p id='headtwo'>Welcome to GC Uni Attendance Portal</p>
    </div>
    <div class="header-right">
      <button class="icon-btn"><i class='bx bx-bell'
id='iconbell'></i></button>
      <button class="icon-btn"><i class='bx bx-cog'
id='iconset'></i></button>
    </div>
  </header>

  <div class="controls-bar">
    <select class='course-select' id="courseSelect" name="Expsn">
      <option value="Null" disabled selected>Select Course</option>
    </select>

    <select class='section-select'>
      <option>Select Section</option>
    </select>

    <div class="toggle-container">
      <label class="toggle-btn active" id="lblRegular">
        Regular
        <input type="radio" name="att_radio_type" value="Regular"
id="radioRegular" checked>
      </label>
      <label class="toggle-btn" id="lblMakeup">
        Makeup
        <input type="radio" name="att_radio_type" value="Makeup"
id="radioMakeup">
      </label>
    </div>

    <div class="update-btn" onclick="fetchStudentsData()">
      <h4 id='headthree' style="margin-right: 5px;">Update</h4>
      <i class='bx bx-refresh' id='iconupdate' style="font-
size:24px;"></i>
    </div>
  </div>

  <div class="grid-container">
    <div class="grid-header">
      <div class="search-mini" style="background:#F4F7FE; padding:8px
15px; border-radius:20px;">
        <i class='bx bx-search' style="color:var(--text-grey);"></i>
        <input type="text" placeholder="Search List"
style="border:none; background:transparent; outline:none; margin-left:5px;">

```

```

        </div>
    </div>
    <div id="myGrid" class="ag-theme-quartz"></div>
</div>
</div>

<div class="right-sidebar">

    <div class="leave-widget" id="leaveWidgetContainer"
style="display:none;">
        <button class="nav-arrow nav-prev" onclick="prevLeave()"><i class='bx
bx-chevron-left'></i></button>
        <div class="leave-card" id="leaveCard">

            </div>
            <button class="nav-arrow nav-next" onclick="nextLeave()"><i class='bx
bx-chevron-right'></i></button>
        </div>

        <div class="summary-widget">
            <div style="display:flex; justify-content:space-between; align-
items:center; margin-bottom:20px;">
                <i class='bx bx-chevron-left'></i>
                <h5 class='datetext'>Today <span
id="currentDateDisplay"></span></h5>
                <i class='bx bx-chevron-right'></i>
            </div>

            <div class="progress-circle" id="progressCircle"><div
class="progress-circle-inner" id="progressText">0%</div></div>

            <div class="stats-row">
                <div class="stat-item">
                    <h5 class='present'>Present</h5>
                    <p id="statPresent" class="stat-present">0</p>
                </div>
                <div class="stat-item">
                    <h5 class='total'>Total</h5>
                    <p id="statTotal">0</p>
                </div>
                <div class="stat-item">
                    <h5>Absent</h5>
                    <p id="statAbsent" style="color:var(--red);">0</p>
                </div>
            </div>

            <div class="info-rows">
                <div class="info-row">
                    <span>Course Code:</span>
                    <span class="info-val" id="displayCourseCode">SE40283</span>
                </div>
                <div class="info-row">

```

```

        <span>Saved Time:</span>
        <span class="info-val">No record found</span>
    </div>
    <div class="info-row">
        <span>Current Time:</span>
        <span class="info-val" id="currentTimeDisplay">12:00 |
12/9/2023</span>
    </div>
</div>

    <button class='btn-main' id="markAttendanceBtn">Mark
Attendance</button>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/ag-grid-community/dist/ag-grid-
community.min.js"></script>

<script>

    const th_regId = "<?php echo $loggedUserId; ?>";
    let course_id = "Null";
    let attendance_type = "Regular";
    let gridApi = null;

    const now = new Date();
    const dateStr = now.toLocaleDateString('en-CA');

    document.getElementById('currentDateDisplay').textContent =
now.toDateString();
    document.getElementById('currentTimeDisplay').textContent =
`${now.getHours()}:${String(now.getMinutes()).padStart(2, '0')} |
${now.toLocaleDateString()}`;

    const columnDefs = [
        {
            headerName: 'Name',
            field: 'stu_name',
            headerCheckboxSelection: true,
            checkboxSelection: true,
            flex: 1.5
        },
        { headerName: 'Registration ID', field: 'stuRegIds', flex: 1 },
        {
            headerName: 'Attendance%',
            field: 'attendance_percentage',
            flex: 1,
            cellStyle: params => {
                if (params.value >= 100) return { color: 'green', fontWeight:
'bold' };
                else if (params.value <= 75) return { color: '#EE5D50',
fontWeight: 'bold' };

```

```

        else return { color: 'black', fontWeight: 'bold' };
    },
    cellRenderer: params => {
        return params.value + '<span style="color: rgb(184, 184,
184)">%</span>';
    }
},
{
    headerName: 'Semester',
    field: 'semester',
    flex: 0.8,
    cellRenderer: params => {
        const rom = {1:"I", 2:"II", 3:"III", 4:"IV", 5:"V", 6:"VI",
7:"VII", 8:"VIII"};
        return rom[params.value] || params.value;
    }
},
{
    headerName: 'Request Status',
    field: 'requestStatus',
    flex: 1,
    cellRenderer: params => {
        let dotClass = 'grey-dot';
        let statusText = params.value || '';
        if (params.value === 'Live') dotClass = 'live-blue-dot';
        else if (params.value === 'Approved') dotClass = 'accepted-
green-dot';
        else if (params.value === 'Rejected') dotClass = 'rejected-
red-dot';

        return `<div style="display:flex; align-items:center;"><span
class="dot ${dotClass}"></span> ${statusText}</div>`;
    }
},
{ headerName: "LeaveId", field: "leave_id", hide: true,
suppressToolPanel: true }
];

const gridOptions = {
    columnDefs: columnDefs,
    rowData: [],
    defaultColDef: {
        flex: 1,
        editable: false,
        sortable: true,
        filter: true,
    },
    rowSelection: 'multiple',
    suppressRowClickSelection: true,
    rowHeight: 50,
    headerHeight: 50,

```

```

        onSelectionChanged: updateSummaryWidget
    };

    function updateSummaryWidget() {
        if (!gridApi) return;
        let total = 0;
        let selected = 0;
        gridApi.forEachNode(() => total++);
        gridApi.getSelectedNodes().forEach(() => selected++);
        const absent = total - selected;
        const pct = total > 0 ? ((selected / total) * 100).toFixed(1) : 0;

        document.getElementById('statPresent').textContent = selected;
        document.getElementById('statTotal').textContent = total;
        document.getElementById('statAbsent').textContent = absent;
        document.getElementById('progressText').textContent = pct + '%';
        document.getElementById('progressCircle').style.background =
            `conic-gradient(var(--primary-color) ${pct}%, #edf2f7 0deg)`;
    }

    function fetchTeacherRegisteredCourses() {
        const url =
            `api.php?action=getTeacherCourses&th_regId=${th_regId}&att_date=${dateStr}`;

        fetch(url)
            .then(res => res.json())
            .then(data => {
                const select = document.getElementById('courseSelect');

                select.innerHTML = '<option value="Null" disabled
selected>Select Course</option>';

                data.forEach(item => {
                    const option = document.createElement('option');
                    option.value = item.courseIDS;

                    option.text = item.courseOption;
                    select.appendChild(option);
                });
            })
            .catch(err => console.error('Error fetching courses:', err));
    }

    function fetchStudentsData() {
        if(course_id === "Null") return;

        console.log("Fetching students for:", course_id);
        document.getElementById('displayCourseCode').textContent = course_id;

        const url =
            `api.php?action=getStudents&course_id=${course_id}&th_regId=${th_regId}&leave_date=${dateStr}`;

```

```

    fetch(url)
      .then(res => res.json())
      .then(data => {
        if(gridApi) {
          gridApi.setGridOption('rowData', data);
          updateSummaryWidget();
        }
      })
      .catch(err => console.error('Error fetching students:', err));
}

function exportTableAsJson() {
  if(!gridApi) return;

  const exportData = [];
  const datetime = new Date();
  const date = datetime.toLocaleDateString('en-CA');
  const time = datetime.toString().split(' ')[0];

  gridApi.forEachNode(node => {
    const isSelected = node.isSelected() ? 'Present' : 'Absent';
    const { requestStatus, stuRegIds, leave_id } = node.data;

    const rowDataWithVariables = {
      course_id,
      th_regId,
      isSelected,
      date,
      time,
      Attendance_type: attendance_type,
      stuRegIds,
      leave_id
    };
    exportData.push(rowDataWithVariables);
  });

  fetch('api.php?action=markAttendance', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ data: exportData })
  })
  .then(response => response.json())
  .then(data => {
    console.log('Success:', data);
    window.alert("Attendance Marked Successfully");
    fetchStudentsData();

    fetchTeacherRegisteredCourses();
  })
  .catch(error => {

```

```

        console.error('Error:', error);
        window.alert("Error marking attendance");
    });
}

document.addEventListener('DOMContentLoaded', () => {

    const gridDiv = document.querySelector('#myGrid');
    gridApi = agGrid.createGrid(gridDiv, gridOptions);

    fetchTeacherRegisteredCourses();

    document.getElementById('courseSelect').addEventListener('change',
(e) => {
        course_id = e.target.value;
        fetchStudentsData();
    });

    const radios = document.getElementsByName('att_radio_type');
    const lblRegular = document.getElementById('lblRegular');
    const lblMakeup = document.getElementById('lblMakeup');

    radios.forEach(radio => {
        radio.addEventListener('change', (e) => {
            attendance_type = e.target.value;

            if(attendance_type === 'Regular') {
                lblRegular.classList.add('active');
                lblMakeup.classList.remove('active');
            } else {
                lblMakeup.classList.add('active');
                lblRegular.classList.remove('active');
            }
        });
    });

    document.getElementById('markAttendanceBtn').addEventListener('click'
, exportTableAsJson);

    fetchPendingLeaves();
});

let pendingLeaves = [];
let currentLeaveIndex = 0;

function fetchPendingLeaves() {
    fetch(`api.php?action=getPendingLeaves&th_regId=${th_regId}`)
        .then(res => res.json())
        .then(data => {
            pendingLeaves = data;
            currentLeaveIndex = 0;
            renderLeaveCard();
        });
}

```

```

    })
    .catch(err => console.error(err));
}

function renderLeaveCard() {
    const container = document.getElementById('leaveWidgetContainer');
    if (!pendingLeaves || pendingLeaves.length === 0) {
        container.style.display = 'none';
        return;
    }
    container.style.display = 'flex';

    const leave = pendingLeaves[currentLeaveIndex];
    const card = document.getElementById('leaveCard');

    card.innerHTML = `
        <div class="leave-header">
            <div>
                <div class="leave-stu-name">${leave.stu_name}</div>
                <div class="leave-id-badge">${leave.stu_regId}</div>
            </div>
            <div class="leave-title">${leave.leave_title}</div>
        </div>
        <div class="leave-message">
            ${leave.leave_message}
        </div>
        <div class="leave-footer">
            <div class="leave-date">${leave.leave_date}</div>
            <div class="leave-actions">
                <button class="btn-accept"
onclick="updateLeaveStatus(${leave.leave_id}, 'Approved')">Accept</button>
                <button class="btn-reject"
onclick="updateLeaveStatus(${leave.leave_id}, 'Rejected')">Reject</button>
            </div>
        </div>
    `;
}

function nextLeave() {
    if (pendingLeaves.length === 0) return;
    currentLeaveIndex = (currentLeaveIndex + 1) % pendingLeaves.length;
    renderLeaveCard();
}

function prevLeave() {
    if (pendingLeaves.length === 0) return;
    currentLeaveIndex = (currentLeaveIndex - 1 + pendingLeaves.length) %
pendingLeaves.length;
    renderLeaveCard();
}

function updateLeaveStatus(id, status) {

```

```
    fetch('api.php?action=updateLeaveStatus', {
      method: 'POST',
      body: JSON.stringify({ leave_id: id, status: status })
    })
    .then(res => res.json())
    .then(data => {
      pendingLeaves.splice(currentLeaveIndex, 1);
      if (currentLeaveIndex >= pendingLeaves.length) {
        currentLeaveIndex = 0;
      }
      renderLeaveCard();
      fetchStudentsData();
    })
    .catch(err => console.error(err));
  }

</script>
</body>
</html>
```

login.php

```
<?php
session_start();

$loginStatus = "";
$statusHolder = "hideMessage";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $servername = "localhost";
    $username = "root";
    $password = "";
    $dbname = "attendance_portal_beenishtasaffar";

    $conn = new mysqli($servername, $username, $password, $dbname);

    if ($conn->connect_error) {
        die("Connection failed: " . $conn->connect_error);
    }

    $loginUserName = $_POST['loginUserName'];
    $loginPassword = $_POST['loginPassword'];

    $stmt = $conn->prepare("SELECT th_regId FROM login WHERE l_username = ? AND
l_password = ?");
    $stmt->bind_param("ss", $loginUserName, $loginPassword);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows > 0) {
        $row = $result->fetch_assoc();

        $_SESSION['isLoggedIn'] = true;
        $_SESSION['LoggedInUserId'] = $row['th_regId'];

        header("Location: portal.php");
        exit();
    } else {

        $loginStatus = "Credentials Don't Exist";
        $statusHolder = "showMessage";
    }

    $stmt->close();
    $conn->close();
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
```

```

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login - AU Attendance Portal</title>

<link rel="stylesheet" href=".\\css\\login.css">
<style>
    body { margin: 0; padding: 0; font-family: 'Manrope', sans-serif; }
    a { text-decoration: none; color: inherit; }
</style>
</head>
<body>
    <div>
        <div class="background-image-div">
            
        </div>
        <div class="login-container">
            <div class="grey-box">
                <div class="login-content">
                    <div class="one">
                        <h1 class="login-heading">Login</h1>
                        <h5 class="au-heading">Welcome to GC Attendance Portal</h5>
                    </div>

                    <span class="<?php echo $statusHolder; ?>"><?php echo $loginStatus;
?></span>

                    <div class="form">
                        <form action="login.php" method="POST" class="login-form">
                            <div class="fields">
                                <input type="text" name="loginUserName" id="username"
class="login-input1" placeholder="    Email" required />
                                <input type="password" name="loginPassword" id="password"
class="login-input2" placeholder="    Password" required />
                            </div>
                            <div class="fieldBottomLines">

                                <div class="rememberMeDiv">
                                    <input type="checkbox" id="rememberMe" name="rememberMe"
class="remember-me-checkbox" />
                                    <label for="rememberMe" class="remember-me-label"></label>
                                    <label for="rememberMe" class="remember-me-text">Remember
me</label>
                                </div>

                                <a href="#" class="forget-password-link">Forget Password?</a>
                                </div>
                            <div class="login-section">
                                <div class="login-button-section">
                                    <button type="submit" class="login-button"> Login </button>
                                </div>
                                <h5 class="signin-heading">or sign in with</h5>

```

```

        </div>
    </form>
</div>

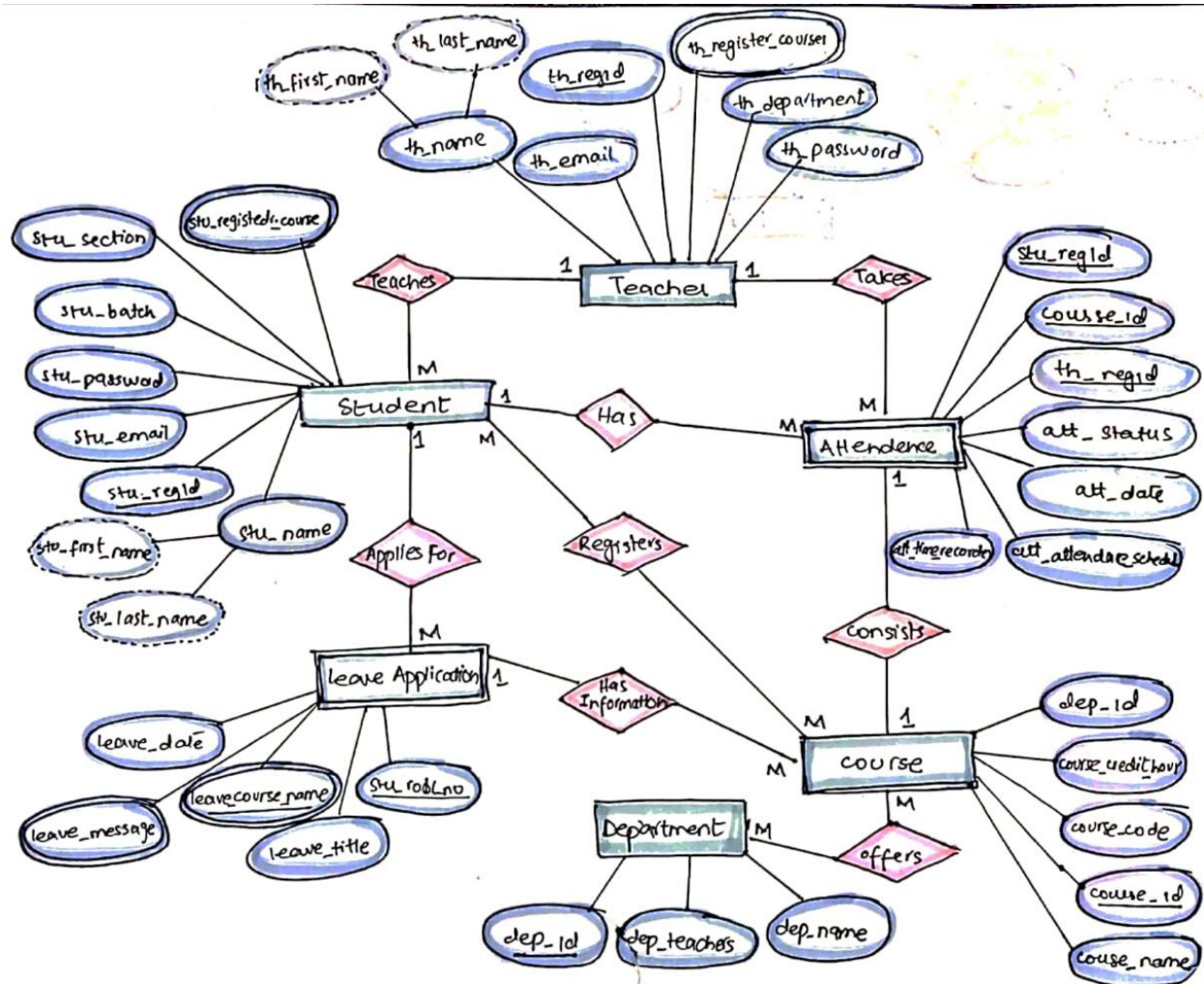
<div class="last">
    <div class="loginOptions">
        <button type="button" class="google-logo-button">
            
        </button>
        <button type="button" class="uni-logo-button">
            
        </button>
    </div>
    <div class="lastBox">
        <p class="last-heading"> Dont have an account?</p>
        <a href="register.php" class="signup"> <span style="font-weight:
700; text-decoration: underline;"> Sign up </span> here </a>
    </div>
</div>
</div>
</div>
</div>
</div>

<script>
    document.addEventListener("DOMContentLoaded", function() {
        const statusSpan = document.querySelector('.showMessage');
        if (statusSpan && statusSpan.textContent.trim() !== "") {
            setTimeout(() => {
                statusSpan.classList.remove('showMessage');
                statusSpan.classList.add('hideMessage');
            }, 4000);
        }
    });
</script>
</body>
</html>

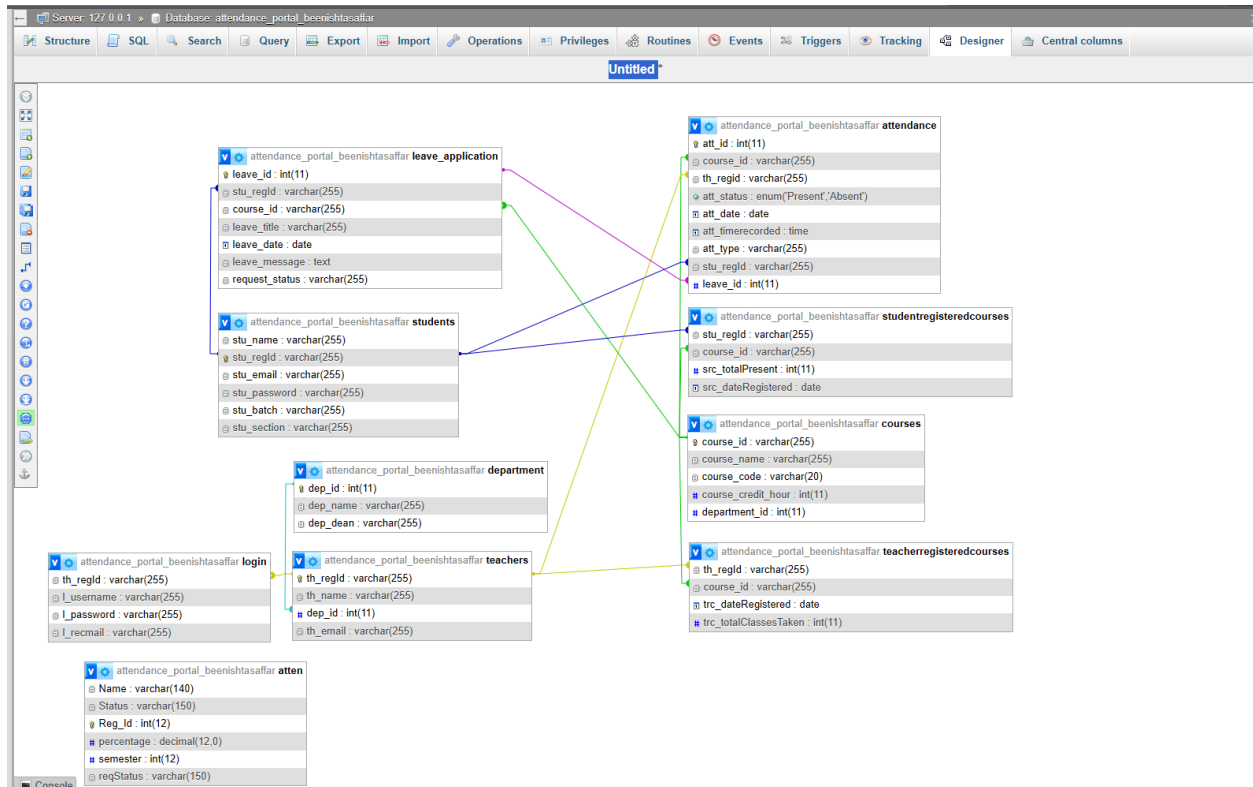
```

10. Database Structure:

Entity Relationship Diagram:



Relationships From XAMPP Interface:



7. Database Queries

- ```
SELECT th_regId FROM login WHERE l_username = ? && l_password = ?;
```
- ```
SELECT
    students.stu_name,
    sub_query.IDs AS stuRegIds,
    sub_query.attendance_percentage,
    2026 - students.stu_batch AS semester,
    sub_query.RQs AS requestStatus,
    sub_query.lids AS leave_id
FROM students
LEFT JOIN (
    SELECT
        studentregisteredcourses.stu_regId AS IDs,
        ROUND((COALESCE(studentregisteredcourses.src_totalPresent, 0) / COALESCE(teacherregisteredcourses.trc_totalClassesTaken, 1)) * 100, 2) AS attendance_percentage,
        COALESCE(leave_application.request_status, 'No Request') AS RQs,
        COALESCE(leave_application.leave_id) AS lids
    FROM studentregisteredcourses
    INNER JOIN students ON studentregisteredcourses.stu_regId = students.stu_regId
    LEFT JOIN leave_application ON students.stu_regId = leave_application.stu_regId
    AND leave_application.leave_date = ?
    LEFT JOIN teacherregisteredcourses ON studentregisteredcourses.course_id = teacherregisteredcourses.course_id
    AND teacherregisteredcourses.th_regId = ?
    AND teacherregisteredcourses.course_id = ?
    WHERE studentregisteredcourses.course_id = ?
) AS sub_query ON students.stu_regId = sub_query.IDs
WHERE sub_query.IDs IS NOT NULL;
```
- ```
SELECT courses.course_id AS courseIDS, CONCAT(courses.course_id, ' - ', courses.course_name) AS courseOption
FROM (
 SELECT trc.course_id, crs.course_name
 FROM teacherregisteredcourses trc
 JOIN courses crs ON trc.course_id = crs.course_id
 WHERE trc.th_regId = ?
) AS courses
LEFT JOIN attendance at ON courses.course_id = at.course_id AND at.att_date = ?
WHERE at.course_id IS NULL
.
```
- ```
INSERT INTO login(th_regId, l_username, l_password, l_recmail) VALUES (?, ?, ?, ?);
```
- ```
INSERT INTO attendance (course_id, th_regid, att_status, att_date, att_timerecorded, att_type, stu_regId, leave_id)
VALUES (?);
```
- ```
UPDATE teacherregisteredcourses
    SET trc_totalClassesTaken = trc_totalClassesTaken + 1
    WHERE th_regId = "${th_regId}"
    AND course_id = "${course_id}";
```
- ```
UPDATE studentregisteredcourses
 SET src_totalPresent = src_totalPresent + 1
 WHERE stu_RegId = "${stuRegIds}"
 AND course_id = "${course_id}";
```

## 1. User Registration Query

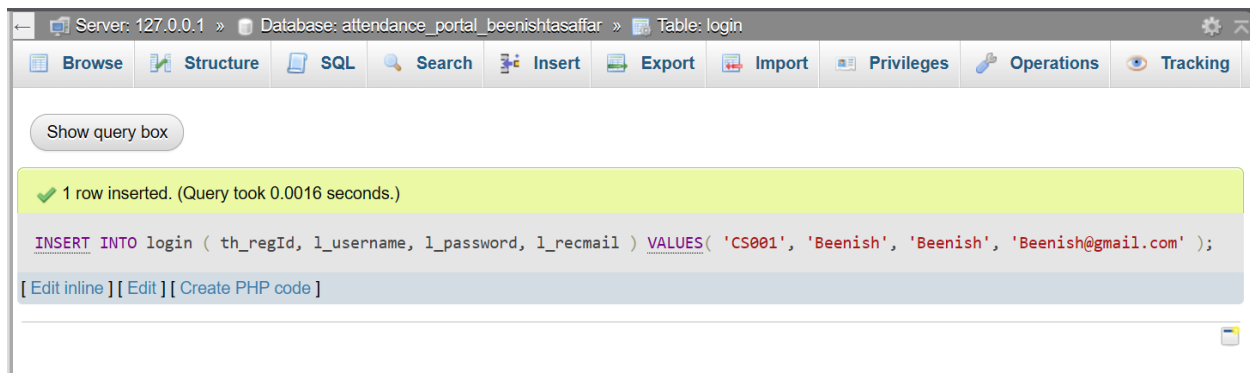
### Generalized Query

```
INSERT INTO login (
 th_regId,
 l_username,
 l_password,
 l_recmail
)
VALUES (?, ?, ?, ?);
```

### Example Query

```
INSERT INTO login (
 th_regId,
 l_username,
 l_password,
 l_recmail
)
VALUES(
 'CS001',
 'Beenish',
 'Beenish',
 'Beenish@gmail.com'
);
```

### Screenshot of Example Query



## 2. User Login Query

### Generalized Query

```
SELECT th_regId
FROM login
WHERE l_username = ?
AND l_password = ?;
```

### Example Query

```
SELECT th_regId
FROM login
WHERE l_username = 'Beenish'
AND l_password = 'Beenish';
```

### Screenshot of Example Query

The screenshot displays a database management interface with the following components:

- Header:** Server: 127.0.0.1 » Database: attendance\_portal\_beenishtasaffar » Table: login
- Navigation Bar:** Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, More
- Status Bar:** Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)
- Query Editor:** `SELECT th_regId FROM login WHERE l_username = 'Beenish' AND l_password = 'Beenish';`
- Tools:** Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]
- Filters:** Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None
- Extra options:** th\_regId
- Results:** CS001, CS001

### 3. Student Attendance Dashboard Query

#### *Generalized Query*

```
SELECT
 s.stu_name,
 src.stu_regId,
 (src.src_totalPresent / trc.trc_totalClassesTaken) * 100 AS attendance_percentage,
 2026 - s.stu_batch AS semester,
 COALESCE(la.request_status, 'No Request') AS requestStatus
FROM studentregisteredcourses src
JOIN students s ON s.stu_regId = src.stu_regId
JOIN teacherregisteredcourses trc
 ON trc.course_id = src.course_id
 AND trc.th_regId = ?
LEFT JOIN leave_application la
 ON la.stu_regId = s.stu_regId
 AND la.course_id = src.course_id
 AND la.leave_date = ?
WHERE src.course_id = ?;
```

#### *Example Query*

```
SELECT
 s.stu_name,
 src.stu_regId,
 ROUND((src.src_totalPresent / trc.trc_totalClassesTaken) * 100, 2) AS attendance_percentage,
 2026 - s.stu_batch AS semester,
 COALESCE(la.request_status, 'No Request') AS requestStatus
FROM studentregisteredcourses src
JOIN students s ON s.stu_regId = src.stu_regId
JOIN teacherregisteredcourses trc
 ON trc.course_id = src.course_id
 AND trc.th_regId = 'CS001'
LEFT JOIN leave_application la
 ON la.stu_regId = s.stu_regId
 AND la.course_id = src.course_id
 AND la.leave_date = '2026-05-17'
WHERE src.course_id = 'CSE101';
```

## Screenshot of Example Query

Server: 127.0.0.1 » Database: attendance\_portal\_beenishtasaffar » Table: studentregisteredcourses

[Browse](#) [Structure](#) [SQL](#) [Search](#) [Insert](#) [Export](#) [Import](#) [Privileges](#) [Operations](#) [More](#)

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 2 (3 total, Query took 0.0096 seconds.)

```
SELECT s.stu_name, src.stu_regId, ROUND((src.src_totalPresent / trc.trc_totalClassesTaken) * 100, 2) AS attendance_percentage,
2026 - s.stu_batch AS semester, COALESCE(la.request_status, 'No Request') AS requestStatus FROM studentregisteredcourses src
JOIN students s ON s.stu_regId = src.stu_regId JOIN teacherregisteredcourses trc ON trc.course_id = src.course_id AND
trc.th_regId = 'CS001' LEFT JOIN leave_application la ON la.stu_regId = s.stu_regId AND la.course_id = src.course_id AND
la.leave_date = '2026-05-17' WHERE src.course_id = 'CSE101';
```

☐ Profiling [\[ Edit inline \]](#) [\[ Edit \]](#) [\[ Explain SQL \]](#) [\[ Create PHP code \]](#) [\[ Refresh \]](#)

☐ Show all | Number of rows: 25 ▼ Filter rows:  Sort by key: None

Extra options

| stu_name         | stu_regId | attendance_percentage | semester | requestStatus |
|------------------|-----------|-----------------------|----------|---------------|
| Ahmad Hassan     | 221775    | 34.38                 | 0        | Approved      |
| Beenish Tasaffar | 221847    | 34.38                 | 0        | No Request    |
| Faraz Ashraf     | 221899    | 37.50                 | 0        | No Request    |

☐ Show all | Number of rows: 25 ▼ Filter rows:  Sort by key: None

#### 4. Teacher Available Courses Query

##### *Generalized Query*

```
SELECT
 c.course_id,
 CONCAT(c.course_id, '-', c.course_name)
FROM teacherregisteredcourses trc
JOIN courses c ON c.course_id = trc.course_id
LEFT JOIN attendance a
 ON a.course_id = trc.course_id
 AND a.att_date = ?
WHERE trc.th_regId = ?
AND a.course_id IS NULL;
```

##### *Example Query*

```
SELECT
 c.course_id,
 CONCAT(c.course_id, '-', c.course_name)
FROM teacherregisteredcourses trc
JOIN courses c ON c.course_id = trc.course_id
LEFT JOIN attendance a
 ON a.course_id = trc.course_id
 AND a.att_date = '2026-05-7'
WHERE trc.th_regId = 'CS001'
AND a.course_id IS NULL;
```

## Screenshot of Example Query

The screenshot shows a database management interface with a top navigation bar containing buttons for Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and More. Below the navigation bar, a status bar indicates "Showing rows 0 - 0 (1 total, Query took 0.0015 seconds.)". The main area displays a SQL query: `SELECT c.course_id, CONCAT(c.course_id, ' - ', c.course_name) FROM teacherregisteredcourses trc JOIN courses c ON c.course_id = trc.course_id LEFT JOIN attendance a ON a.course_id = trc.course_id AND a.att_date = '2026-05-7' WHERE trc.th_regId = 'CS001' AND a.course_id IS NULL;`. Below the query, there are links for Profiling, Edit inline, Edit, Explain SQL, Create PHP code, and Refresh. A control bar includes a "Show all" checkbox, a "Number of rows" dropdown set to 25, and a "Filter rows" search box. An "Extra options" button is also present. The results table has two columns: "course\_id" and "CONCAT(c.course\_id, ' - ', c.course\_name)". The first row shows "CSE101" and "CSE101 - Introduction to Programming".

| course_id | CONCAT(c.course_id, ' - ', c.course_name) |
|-----------|-------------------------------------------|
| CSE101    | CSE101 - Introduction to Programming      |

## 5. Insert Attendance Record

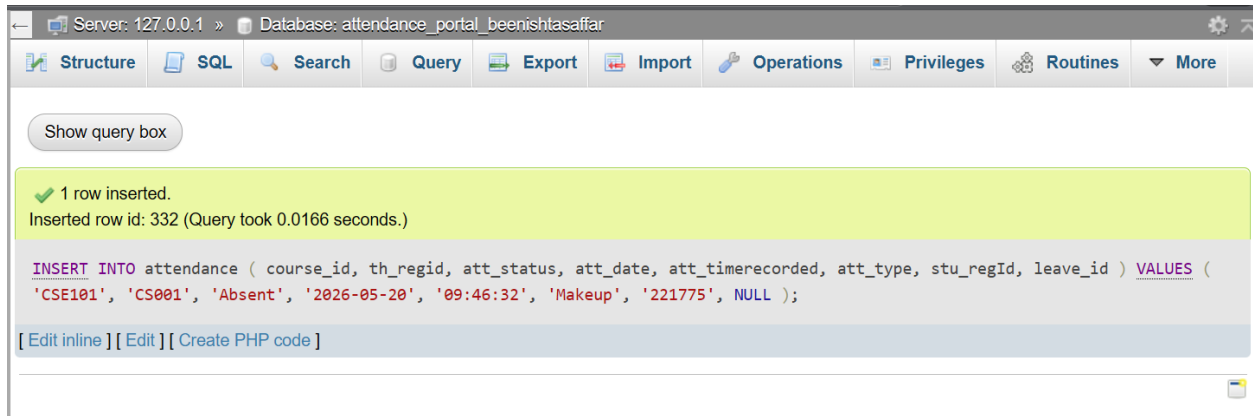
### Generalized Query

```
INSERT INTO attendance (
 course_id, th_regid, att_status, att_date, att_timerecorded,
 att_type, stu_regId, leave_id
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?);
```

### Example Query

```
INSERT INTO attendance (
 course_id, th_regid, att_status, att_date, att_timerecorded,
 att_type, stu_regId, leave_id
)
VALUES (
 'CSE101',
 'CS001',
 'Absent',
 '2026-05-20',
 '09:46:32',
 'Makeup',
 '221775',
 NULL
);
```

## Screenshot of Example Query



## 6. Update Attendance Statistics (Teacher)

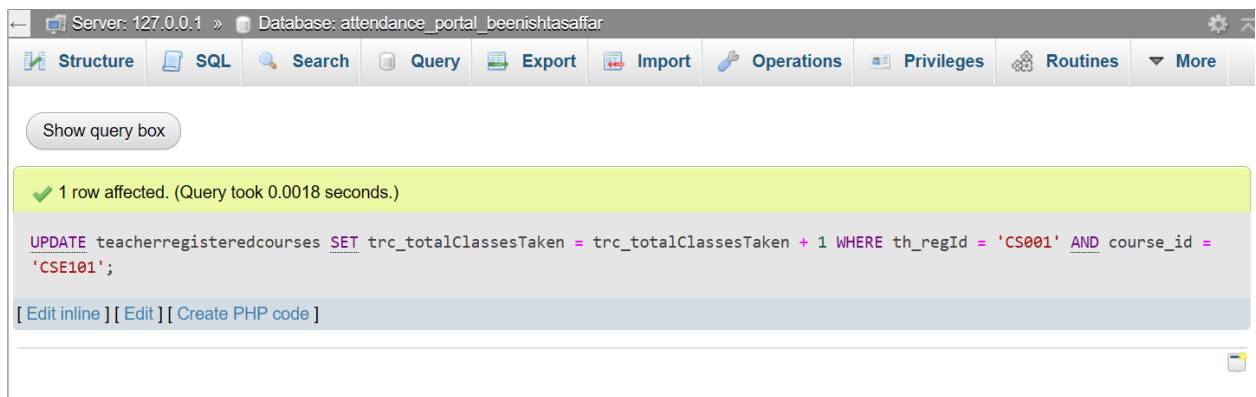
### Generalized Query

```
UPDATE teacherregisteredcourses
SET trc_totalClassesTaken = trc_totalClassesTaken + 1
WHERE th_regId = ?
AND course_id = ?;
```

### Example Query

```
UPDATE teacherregisteredcourses
SET trc_totalClassesTaken = trc_totalClassesTaken + 1
WHERE th_regId = 'CS001'
AND course_id = 'CSE101';
```

### Screenshot of Example Query



## 7. Update Student Total Present Count

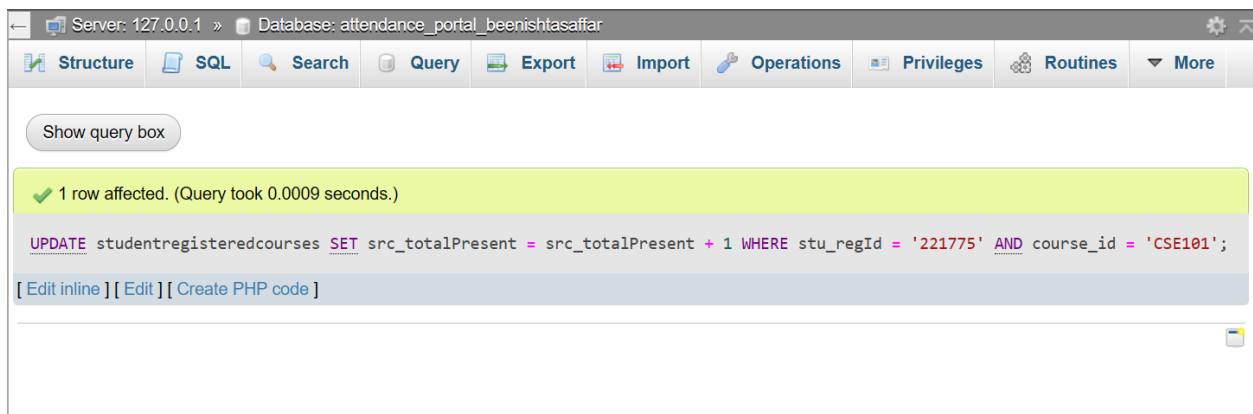
### Generalized Query

```
UPDATE studentregisteredcourses
SET src_totalPresent = src_totalPresent + 1
WHERE stu_regId = ?
AND course_id = ?;
```

### Example Query

```
UPDATE studentregisteredcourses
SET src_totalPresent = src_totalPresent + 1
WHERE stu_regId = '221775'
AND course_id = 'CSE101';
```

### Screenshot of Example Query



## 8. Leave Application Lookup

### Generalized Query

**SELECT**

leave\_id, stu\_regId, course\_id, leave\_title, leave\_date, leave\_message, request\_status

**FROM** leave\_application

**WHERE** stu\_regId = ?

**AND** course\_id = ?;

### Example Query

**SELECT**

leave\_id, stu\_regId, course\_id, leave\_title, leave\_date, leave\_message, request\_status

**FROM** leave\_application

**WHERE** stu\_regId = '221775'

**AND** course\_id = 'CSE101';

### Screenshot of Example Query

Server: 127.0.0.1 » Database: attendance\_portal\_beenishtasaffar » Table: leave\_application

Showing rows 0 - 0 (1 total, Query took 0.0012 seconds.)

```
SELECT leave_id, stu_regId, course_id, leave_title, leave_date, leave_message, request_status FROM leave_application WHERE stu_regId = '221775' AND course_id = 'CSE101';
```

☐ Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                                           | leave_id | stu_regId | course_id | leave_title | leave_date | leave_message                           | request_status |
|-------------------------------------------|----------|-----------|-----------|-------------|------------|-----------------------------------------|----------------|
| <input type="checkbox"/> Edit Copy Delete | 1        | 221775    | CSE101    | Sickness    | 2026-05-17 | Hi, Ma'am. It's me Ahmad, and I'm sick. | Approved       |

Check all With selected: Edit Copy Delete Export

## 8. SQL File:

```
-- phpMyAdmin SQL Dump
-- version 5.2.1
-- https://www.phpmyadmin.net/
--
-- Host: 127.0.0.1
-- Generation Time: Feb 02, 2026 at 11:16 PM
-- Server version: 10.4.32-MariaDB
-- PHP Version: 8.0.30

SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
START TRANSACTION;
SET time_zone = "+00:00";

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;

--
-- Database: `attendance_portal_beenishtasaffar`
--
-- -----
--
-- Table structure for table `atten`
--

CREATE TABLE `atten` (
 `Name` varchar(140) NOT NULL,
 `Status` varchar(150) NOT NULL,
 `Reg_Id` int(12) NOT NULL,
 `percentage` decimal(12,0) NOT NULL,
 `semester` int(12) NOT NULL,
 `reqStatus` varchar(150) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;

--
-- Dumping data for table `atten`
--

INSERT INTO `atten` (`Name`, `Status`, `Reg_Id`, `percentage`, `semester`, `reqStatus`) VALUES
('Faraz Ashraf', 'absent', 22237, 100, 6, 'Live'),
('Voldemort', 'Leave', 221312, 69, 9, 'accepted'),
('Abdullah Hassan', 'Present', 221754, 21, 1, 'accepted'),
('Ahmad Hassan', 'Absent', 221775, 89, 4, 'rejected'),
('Beenish Tasaffar', 'present', 221847, 74, 6, 'present');
```

```
--
-- Table structure for table `attendance`
--
```

```
CREATE TABLE `attendance` (
 `att_id` int(11) NOT NULL,
 `course_id` varchar(255) NOT NULL,
 `th_regid` varchar(255) NOT NULL,
 `att_status` enum('Present','Absent') DEFAULT NULL,
 `att_date` date DEFAULT NULL,
 `att_timerecorded` time DEFAULT NULL,
 `att_type` varchar(255) DEFAULT NULL,
 `stu_regid` varchar(255) NOT NULL,
 `leave_id` int(11) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `attendance`
--
```

```
INSERT INTO `attendance` (`att_id`, `course_id`, `th_regid`, `att_status`, `att_date`, `att_timerecorded`,
 `att_type`, `stu_regid`, `leave_id`) VALUES
(329, 'CSE101', 'CS001', 'Absent', '2026-05-20', '09:46:32', 'Makeup', '221775', NULL),
(330, 'CSE101', 'CS001', 'Absent', '2026-05-20', '09:46:32', 'Makeup', '221847', NULL),
(331, 'CSE101', 'CS001', 'Absent', '2026-05-20', '09:46:32', 'Makeup', '221899', NULL),
(332, 'CSE101', 'CS001', 'Absent', '2026-05-20', '09:46:32', 'Makeup', '221775', NULL);
```

```

--
-- Table structure for table `courses`
--
```

```
CREATE TABLE `courses` (
 `course_id` varchar(255) NOT NULL,
 `course_name` varchar(255) NOT NULL,
 `course_code` varchar(20) NOT NULL,
 `course_credit_hour` int(11) NOT NULL,
 `department_id` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `courses`
--
```

```
INSERT INTO `courses` (`course_id`, `course_name`, `course_code`, `course_credit_hour`, `department_id`)
VALUES
('CSE101', 'Introduction to Programming', 'CS-101', 3, 1),
('CSE220', 'Data Structures and Algorithms', 'CS-220', 3, 1);
```

```

```

```
--
-- Table structure for table `department`
--
```

```
CREATE TABLE `department` (
 `dep_id` int(11) NOT NULL,
 `dep_name` varchar(255) NOT NULL,
 `dep_dean` varchar(255) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `department`
--
```

```
INSERT INTO `department` (`dep_id`, `dep_name`, `dep_dean`) VALUES
(1, 'Computer Science', 'Dr. Ahsan Ali'),
(2, 'Electrical Engineering', 'Dr. Nadia Khan'),
(3, 'Mathematics', 'Dr. Ali Hassan');
```

```

--
-- Table structure for table `leave_application`
--
```

```
CREATE TABLE `leave_application` (
 `leave_id` int(11) NOT NULL,
 `stu_regId` varchar(255) NOT NULL,
 `course_id` varchar(255) NOT NULL,
 `leave_title` varchar(255) NOT NULL,
 `leave_date` date NOT NULL,
 `leave_message` text DEFAULT NULL,
 `request_status` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `leave_application`
--
```

```
INSERT INTO `leave_application` (`leave_id`, `stu_regId`, `course_id`, `leave_title`, `leave_date`,
 `leave_message`, `request_status`) VALUES
(1, '221775', 'CSE101', 'Sickness', '2026-05-17', 'Hi, Ma\`am. It\'s me Ahmad, and I\'m sick.', 'Approved');
```

```

--
-- Table structure for table `login`
--
```

```
CREATE TABLE `login` (
 `th_regId` varchar(255) NOT NULL,
 `l_username` varchar(255) NOT NULL,
```

```

 `l_password` varchar(255) NOT NULL,
 `l_recmail` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;

--
-- Dumping data for table `login`
--
INSERT INTO `login` (`th_regId`, `l_username`, `l_password`, `l_recmail`) VALUES
('CS001', 'ted', 'ted', 'ted@gmail.com'),
('CS001', 'ted', 'ted', 'ted@gmail.com'),
('CS002', 'emial', 'username', 'pass'),
('CS001', 'beenish', 'beenish', 'beenish@gmail.com'),
('CS001', 'Beenish', 'Beenish', 'Beenish@gmail.com');

--
-- Table structure for table `studentregisteredcourses`
--

CREATE TABLE `studentregisteredcourses` (
 `stu_regId` varchar(255) NOT NULL,
 `course_id` varchar(255) NOT NULL,
 `src_totalPresent` int(11) DEFAULT NULL,
 `src_dateRegistered` date DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;

--
-- Dumping data for table `studentregisteredcourses`
--

INSERT INTO `studentregisteredcourses` (`stu_regId`, `course_id`, `src_totalPresent`, `src_dateRegistered`)
VALUES
('221775', 'CSE101', 12, '0000-00-00'),
('221847', 'CSE101', 11, '0000-00-00'),
('221899', 'CSE101', 12, '2026-05-01'),
('221909', 'CSE220', 12, '2026-05-08');

--
-- Table structure for table `students`
--

CREATE TABLE `students` (
 `stu_name` varchar(255) NOT NULL,
 `stu_regId` varchar(255) NOT NULL,
 `stu_email` varchar(255) NOT NULL,
 `stu_password` varchar(255) NOT NULL,
 `stu_batch` varchar(255) NOT NULL,
 `stu_section` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;

```

```
--
-- Dumping data for table `students`
--

INSERT INTO `students` (`stu_name`, `stu_regId`, `stu_email`, `stu_password`, `stu_batch`, `stu_section`)
VALUES
('Ahmad Hassan', '221775', 'ahmad@gmail.com', 'password123', '2026', 'A'),
('Beenish Tasaffar', '221847', 'beenishTasaffar@gmail.com', 'password456', '2026', 'A'),
('Faraz Ashraf', '221899', 'faraz@gmail.com', 'password789', '2026', 'A'),
('Hafiz B', '221909', 'sectionb.com', 'knasdnfkj', '2026', 'B'),
('Hafiz BB', '2219809', 'sklksd@gmailcom', 'pasokdjf', '2026', 'B');
```

```
--
-- Table structure for table `teacherregisteredcourses`
--
```

```
CREATE TABLE `teacherregisteredcourses` (
 `th_regId` varchar(255) NOT NULL,
 `course_id` varchar(255) NOT NULL,
 `trc_dateRegistered` date NOT NULL,
 `trc_totalClassesTaken` int(11) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `teacherregisteredcourses`
--
```

```
INSERT INTO `teacherregisteredcourses` (`th_regId`, `course_id`, `trc_dateRegistered`,
`trc_totalClassesTaken`) VALUES
('CS001', 'CSE101', '2026-05-07', 33);
```

```
--
-- Table structure for table `teachers`
--
```

```
CREATE TABLE `teachers` (
 `th_regId` varchar(255) NOT NULL,
 `th_name` varchar(255) NOT NULL,
 `dep_id` int(11) NOT NULL,
 `th_email` varchar(255) NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci;
```

```
--
-- Dumping data for table `teachers`
--
```

```
INSERT INTO `teachers` (`th_regId`, `th_name`, `dep_id`, `th_email`) VALUES
```

```
('CS001', 'Dr. Michael Smith', 1, 'michael.smith@university.edu'),
('CS002', 'Ms. Sophia Garcia', 1, 'sophia.garcia@university.edu');
```

```
--
-- Indexes for dumped tables
--
```

```
--
-- Indexes for table `atten`
--
```

```
ALTER TABLE `atten`
 ADD PRIMARY KEY (`Reg_Id`);
```

```
--
-- Indexes for table `attendance`
--
```

```
ALTER TABLE `attendance`
 ADD PRIMARY KEY (`att_id`),
 ADD KEY `FK_attendance_course_id` (`course_id`),
 ADD KEY `FK_attendance_th_regid` (`th_regid`),
 ADD KEY `FK_attendance_stu_regid` (`stu_regId`),
 ADD KEY `fk_leave_id` (`leave_id`);
```

```
--
-- Indexes for table `courses`
--
```

```
ALTER TABLE `courses`
 ADD PRIMARY KEY (`course_id`);
```

```
--
-- Indexes for table `department`
--
```

```
ALTER TABLE `department`
 ADD PRIMARY KEY (`dep_id`);
```

```
--
-- Indexes for table `leave_application`
--
```

```
ALTER TABLE `leave_application`
 ADD PRIMARY KEY (`leave_id`),
 ADD KEY `FK_leave_application_course_id` (`course_id`),
 ADD KEY `FK_leave_application_stu_regId` (`stu_regId`);
```

```
--
-- Indexes for table `login`
--
```

```
ALTER TABLE `login`
 ADD KEY `FK_login_th_regid` (`th_regId`);
```

```
--
```

```

-- Indexes for table `studentregisteredcourses`
--
ALTER TABLE `studentregisteredcourses`
 ADD KEY `FK_studentRegisteredCourses_stu_regId` (`stu_regId`),
 ADD KEY `FK_studentRegisteredCourses_course_id` (`course_id`);

--
-- Indexes for table `students`
--
ALTER TABLE `students`
 ADD PRIMARY KEY (`stu_regId`);

--
-- Indexes for table `teacherregisteredcourses`
--
ALTER TABLE `teacherregisteredcourses`
 ADD KEY `th_regId` (`th_regId`),
 ADD KEY `course_id` (`course_id`);

--
-- Indexes for table `teachers`
--
ALTER TABLE `teachers`
 ADD PRIMARY KEY (`th_regId`),
 ADD KEY `FK_teachers_dep_id` (`dep_id`);

--
-- AUTO_INCREMENT for dumped tables
--
--
-- AUTO_INCREMENT for table `attendance`
--
ALTER TABLE `attendance`
 MODIFY `att_id` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=333;

--
-- AUTO_INCREMENT for table `leave_application`
--
ALTER TABLE `leave_application`
 MODIFY `leave_id` int(11) NOT NULL AUTO_INCREMENT, AUTO_INCREMENT=2;

--
-- Constraints for dumped tables
--
--
-- Constraints for table `attendance`
--
ALTER TABLE `attendance`

```

```

 ADD CONSTRAINT `FK_attendance_course_id` FOREIGN KEY (`course_id`) REFERENCES `courses`
(`course_id`),
 ADD CONSTRAINT `FK_attendance_stu_regId` FOREIGN KEY (`stu_regId`) REFERENCES `students`
(`stu_regId`),
 ADD CONSTRAINT `FK_attendance_th_regId` FOREIGN KEY (`th_regId`) REFERENCES `teachers` (`th_regId`),
 ADD CONSTRAINT `fk_leave_id` FOREIGN KEY (`leave_id`) REFERENCES `leave_application` (`leave_id`) ON
DELETE SET NULL;
--
-- Constraints for table `leave_application`
--
ALTER TABLE `leave_application`
 ADD CONSTRAINT `FK_leave_application_course_id` FOREIGN KEY (`course_id`) REFERENCES `courses`
(`course_id`),
 ADD CONSTRAINT `FK_leave_application_stu_regId` FOREIGN KEY (`stu_regId`) REFERENCES `students`
(`stu_regId`);
--
-- Constraints for table `login`
--
ALTER TABLE `login`
 ADD CONSTRAINT `FK_login_th_regId` FOREIGN KEY (`th_regId`) REFERENCES `teachers` (`th_regId`);
--
-- Constraints for table `studentregisteredcourses`
--
ALTER TABLE `studentregisteredcourses`
 ADD CONSTRAINT `FK_studentRegisteredCourses_course_id` FOREIGN KEY (`course_id`) REFERENCES
`courses` (`course_id`),
 ADD CONSTRAINT `FK_studentRegisteredCourses_stu_regId` FOREIGN KEY (`stu_regId`) REFERENCES
`students` (`stu_regId`);
--
-- Constraints for table `teacherregisteredcourses`
--
ALTER TABLE `teacherregisteredcourses`
 ADD CONSTRAINT `teacherregisteredcourses_ibfk_1` FOREIGN KEY (`th_regId`) REFERENCES `teachers`
(`th_regId`),
 ADD CONSTRAINT `teacherregisteredcourses_ibfk_2` FOREIGN KEY (`course_id`) REFERENCES `courses`
(`course_id`);
--
-- Constraints for table `teachers`
--
ALTER TABLE `teachers`
 ADD CONSTRAINT `FK_teachers_dep_id` FOREIGN KEY (`dep_id`) REFERENCES `department` (`dep_id`);
COMMIT;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;

```